

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Worked Solutions

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \cos\theta$: $y = -t^2 + t + 2 = -(t - \frac{1}{2})^2 + \frac{9}{4}$.
2. Check the vertex and the endpoints $t = -1, 1$, then return to θ .

Final answer

$$\max y = \frac{9}{4} \text{ at } \theta = \frac{\pi}{3}, \frac{5\pi}{3}; \quad \min y = 0 \text{ at } \theta = \pi$$

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \sin\theta$: $y = 1 - t^2 - t = -(t + \frac{1}{2})^2 + \frac{5}{4}$.
2. The endpoint $t = 1$ gives the minimum -1 .

Final answer

$$\max y = \frac{5}{4} \text{ at } \theta = \frac{7\pi}{6}, \frac{11\pi}{6}; \quad \min y = -1 \text{ at } \theta = \frac{\pi}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q003 Written

Solve the four source maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the source challenge $y = \cos^2 \theta - \sin \theta$.

Solution

1. Choose the single trig variable that makes the expression quadratic.
2. The vertex is valid only when it lies in $[-1, 1]$; convert every extremal t back to the corresponding angles.

Final answer

See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.

Worked Solutions

Advanced Theorems and Applications

Q004 Written

Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.

Solution

1. Rewrite $75^\circ = 45^\circ + 30^\circ$ and $\pi/12 = 45^\circ - 30^\circ$.
2. Apply the sine-sum and cosine-difference identities.

Final answer

$$\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos \beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha + \beta)$ and $\cos(\alpha - \beta)$.

Solution

1. The quadrant signs give $\cos \alpha = -1/2$ and $\sin \beta = -4/5$.
2. Substitute in the addition identities without a calculator.

Final answer

$$\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}; \quad \cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$$

Worked Solutions

Advanced Theorems and Applications

Q006 Written

Complete the source addition-theorem exercises *exact values, quadrant – controlled α, β , and the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Solution

1. Convert radians to degrees only when it clarifies the reference angle.
2. Recover the missing sine/cosine value from the quadrant before applying the sum or difference identity.

Final answer

See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.

Q007 Written

Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

Solution

1. Use $105^\circ = 45^\circ + 60^\circ$ and the tangent addition theorem.
2. For the lines, use $m_1 = 2, m_2 = -3$ in $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

Final answer

$$\tan 105^\circ = -2 + \sqrt{3}; \quad \theta = \frac{\pi}{4}$$

Worked Solutions

Advanced Theorems and Applications

Q008 Written

Find the angle between each source pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Solution

1. Convert each line to slope-intercept form.
2. Use the acute angle convention $0 < \theta < \pi/2$, replacing θ_1 by $\pi - \theta_1$ when necessary.

Final answer

The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Worked Solutions

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all source line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Solution

1. Solve the two linear equations produced by the tangent formula for the unknown slope.
2. Use the y -intercepts and the line intersection to obtain each triangle area.

Final answer

See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Worked Solutions

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ **with** α **in** QII, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The quadrant gives $\cos \alpha = -2\sqrt{2}/3$.
2. Apply $\sin 2\alpha = 2 \sin \alpha \cos \alpha$, $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$, then divide.

Final answer

$$\sin 2\alpha = -\frac{4\sqrt{2}}{9}, \quad \cos 2\alpha = \frac{7}{9}, \quad \tan 2\alpha = -\frac{4\sqrt{2}}{7}$$

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The interval places α in QIII, so $\cos \alpha = -\sqrt{5}/3$.
2. Substitute into the three double-angle formulas.

Final answer

$$\sin 2\alpha = \frac{4\sqrt{5}}{9}, \quad \cos 2\alpha = \frac{1}{9}, \quad \tan 2\alpha = 4\sqrt{5}$$

Worked Solutions

Advanced Theorems and Applications

Q012 Written

Solve the source double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Solution

1. Determine the missing trig value from the interval before using the identities.
2. For 3θ , use $4\cos^3\theta - 3\cos\theta$; for 2θ , use the tangent double-angle identity.

Final answer

For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 Written

Given $\cos \alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

Solution

1. Use the half-angle formulas and choose signs from $0 < \alpha/2 < \pi/2$.
2. For $(5\pi/8)$, use $\cos(\pi - 3\pi/8)$.

Final answer

$$\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}, \quad \cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}, \quad \tan \frac{\alpha}{2} = \frac{1}{2}, \quad \cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q014 Written

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Solution

1. Since $\alpha/2$ is in QI, both half-angle roots are positive.
2. Use $3\pi/8 = \pi/2 - \pi/8$ or the half-angle identity.

Final answer

$$\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}, \quad \cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}, \quad \tan \frac{\alpha}{2} = \sqrt{2}, \quad \sin \frac{3\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$$

Q015 Written

Solve the source half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}$, $4\pi < A < 5\pi$,
 $\cos B = -\frac{\sqrt{5}}{3}$, $5\pi < B < 6\pi$.

Solution

1. Halve the interval first; its quadrant controls every sign.
2. Apply the half-angle formula again for $(A/4)$ after finding $(\cos(A/2))$.

Final answer

Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.

Worked Solutions

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Use $\sin 2\theta = 2 \sin \theta \cos \theta$ and factor.
2. Use $\cos 2\theta = 2 \cos^2 \theta - 1$, solve the quadratic in $c = \cos \theta$, and map the interval.

Final answer

$$a \ \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}; \quad b \ \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

Q017 Written

Solve the source equation $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Solution

1. Factor $\cos \theta (2 \sin \theta + 1) = 0$.
2. List the two cosine zeros and the two angles with $\sin \theta = -1/2$.

Final answer

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Q018 Written

Solve all source double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Choose the identity that leaves one trig function only.
2. For inequalities, use a sign chart and include or exclude boundary points exactly as stated.

Final answer

See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Solution

1. Plot $P(1, -\sqrt{3})$, find $r = OP$, then read the angle α .

Final answer

$$r = 2, \quad \alpha = -\frac{\pi}{3}, \quad \sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$$

Worked Solutions

Advanced Theorems and Applications

Q020 Written

Rewrite the source expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in harmonic-addition form.

Solution

1. For $a \sin \theta + b \cos \theta$, use $r = \sqrt{a^2 + b^2}$, $\cos \alpha = a/r$, $\sin \alpha = b/r$.

Final answer

$$\sqrt{2} \sin \theta - \frac{\pi}{4}, \sqrt{2} \sin \theta + \frac{\pi}{4}, 2\sqrt{3} \sin \theta - \frac{\pi}{6}, \sqrt{10} \sin \theta - \arctan \frac{1}{3}$$

Q021 Written

Rewrite all source harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Solution

1. Keep the sign of the cosine coefficient when locating P(a,b).
2. Substitute the obtained (r,a) and expand once to verify both coefficients.

Final answer

See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Worked Solutions

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.**Solution**

1. Synthesize each left-hand side and account for the shifted angle range.
2. The maximum of $\sin \theta + \cos \theta$ is $\sqrt{2} < 3/2$, so the second inequality is always true.

Final answer

$$a \theta = \frac{\pi}{2}, \pi; \quad b \ 0 \leq \theta < 2\pi$$

Q023 Written

Solve $\sin \theta \leq \cos \theta$ on $0 \leq \theta < 2\pi$.**Solution**

1. Rewrite as $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.
2. Read the accepted half-turns and retain the equality endpoints.

Final answer

$$0 \leq \theta \leq \frac{\pi}{4} \quad \text{or} \quad \frac{5\pi}{4} \leq \theta < 2\pi$$

Worked Solutions

Advanced Theorems and Applications

Q024 Written

Solve all source harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Solution

1. After synthesis set $t = \theta + \alpha$ and write the exact t-range.
2. Use one full revolution in t, then subtract α and intersect with the original domain.

Final answer

See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Solution

1. Use $-1 \leq \sin(t) \leq 1$ for the first function.
2. For the second, set $t = \cos \theta$ and complete the square after the double-angle substitution.

Final answer

First: $\max = 2$, $\min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Worked Solutions

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the source functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Solution

1. Reduce double angles or $\sin^2 \theta$ to one variable.
2. Check the vertex and both endpoints of the allowed interval.

Final answer

For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos \theta$ and the endpoint/vertex test.

Q027 Written

Solve all source maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Solution

1. Use the same transformation rules as for equations and inequalities.
2. Never report a vertex outside $[-1, 1]$; compare it with both endpoints.

Final answer

See the numbered solution: synthesize $a \sin \theta + b \cos \theta$ or substitute $t = \cos \theta$, then test the full allowed range.

Worked Solutions

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:**Solution**1. Use $\sin^2 \theta = 1 - t^2$.

Correct option: B

Final answer $t = \cos \theta$

Q029 MCQ

The maximum of $\sin^2 \theta + \cos \theta + 1$ is:**Solution**

1. Complete the square.

Correct option: C

Final answer $9/4$

Worked Solutions

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Solution**

1. Trig sine and cosine values are bounded.

Correct option: B**Final answer** $[-1, 1]$

Q031 MCQ

The identity for $\sin(a+b)$ is:**Solution**

1. Use the addition theorem.

Correct option: A**Final answer** $\sin a \cos b + \cos a \sin b$

Worked Solutions

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Solution**

1. Rewrite 75 as 45+30.

Correct option: B

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Q033 MCQ

A missing trig value is selected using:**Solution**

1. The quadrant fixes the sign.

Correct option: A

Final answer

the quadrant

Worked Solutions

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Solution**

1. The coefficient of x is the slope.

Correct option: B**Final answer** m

Q035 MCQ

The angle formula uses denominator:**Solution**

1. Use the tangent difference theorem.

Correct option: A**Final answer** $1 + m_1 m_2$

Worked Solutions

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Solution

1. Choose the acute representative.

Correct option: B

Final answer

$\pi - \theta_1$

Q037 MCQ

$\sin(2a)$ equals:

Solution

1. Double-angle identity.

Correct option: B

Final answer

$2 \sin a \cos a$

Worked Solutions

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2a)$ can be written as:**Solution**

1. Use the standard formula.

Correct option: A**Final answer**

$$\cos^2 a - \sin^2 a$$

Q039 MCQ

For theta in QIV, sin theta is:**Solution**

1. Read the quadrant.

Correct option: B**Final answer**

negative

Worked Solutions

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Solution**1. Choose the sign from $a/2$.

Correct option: A

Final answer $\sqrt{(1-\cos a)/2}$

Q041 MCQ

The sign of a half-angle root is chosen from:**Solution**

1. Halving changes the interval.

Correct option: A

Final answer

the original quadrant interval

Worked Solutions

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Solution

1. $a/2$ is in QI.

Correct option: B

Final answer

positive

Q043 MCQ

For double-angle equations, first rewrite in:

Solution

1. Use a double-angle identity.

Correct option: A

Final answer

one trig function

Worked Solutions

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Solution**1. Factor after substituting $\sin(2\theta)$.

Correct option: A

Final answer $\sin\theta(2\cos\theta + 1)$

Q045 MCQ

A tangent asymptote is always:**Solution**

1. Tangent is undefined there.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q046 MCQ

For a $\sin \theta + b \cos \theta$, r is:**Solution**

1. It is the length OP.

Correct option: B**Final answer**

$$\sqrt{a^2 + b^2}$$

Q047 MCQ

The harmonic target form is:**Solution**

1. Match the coefficients.

Correct option: A**Final answer**

$$r \sin(\theta + a)$$

Worked Solutions

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Solution**

1. The coefficient point determines direction.

Correct option: A

Final answer $P(a, b)$

Q049 MCQ

After synthesis, t is usually:**Solution**

1. Track the shifted range.

Correct option: A

Final answer $\theta + a$

Worked Solutions

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < 3/2$ is:

Solution

1. $\sqrt{2} < 3/2$.

Correct option: B

Final answer

always true

Q051 MCQ

In a strict inequality, equality endpoints are:

Solution

1. Use open endpoints.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Solution**

1. Multiply the sine range by 2.

Correct option: B

Final answer $[-2, 2]$

Q053 MCQ

For a double-angle maximum problem, use:**Solution**

1. Reduce to $t = \sin \theta$ or $\cos \theta$.

Correct option: A

Final answer

the double-angle formula then a bounded variable

Worked Solutions

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Solution**

1. It is not attainable by sine/cosine.

Correct option: B**Final answer**

rejected

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:**Solution**

1. Use the sine range.

Correct option: D**Final answer**

2

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The addition-theorem value $\sin 75^\circ$ is:

Solution

1. Use 45 degrees plus 30 degrees.

Correct option: A

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Solution

1. Complete the square.

Correct option: A

Final answer

$$-2$$

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Solution

1. The tangent magnitude is 1.

Correct option: B

Final answer

$$\pi/4$$

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Solution

1. Let $t = \cos \theta$ and complete the square.

Final answer

$$\max = \frac{9}{4}, \min = 0$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Solution

1. Use the quadrant to select $\cos \alpha$.

Final answer

$$-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$$

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Solution

1. Find r and the coefficient angle from $P(3, -\sqrt{3})$.

Final answer

$$2\sqrt{3} \sin \theta - \frac{\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 8 **Concept Quiz · Written****Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.**Solution**1. Synthesize to $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.**Final answer**

$$[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$$

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge,
 $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$,
 $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{2}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a \theta = \frac{\pi}{2}, \pi$; $b \ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q025** First: $\max = 2$, $\min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q026** For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos \theta$ and the endpoint/vertex test.
- Q027** See the numbered solution: synthesize $a \sin \theta + b \cos \theta$ or substitute $t = \cos \theta$, then test the full allowed range.
- Q028** B
- Q029** C
- Q030** B
- Q031** A
- Q032** B
- Q033** A
- Q034** B
- Q035** A
- Q036** B
- Q037** B
- Q038** A
- Q039** B
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}$, $\min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$